

QUALCOMM SNAPDRAGON X HACKATHON

DUAL-AI ON-DEVICE | ZERO CLOUD

MIRROR

The Privacy-First, On-Device Digital Wellbeing Companion

Unleashing Qualcomm Hexagon NPU & Qualcomm AI Hub on Windows 11 Copilot+ PCs

HEXAGON NPU ACCELERATION

Continuous [60x10] sequence classifier running via QNN EP in 1.42ms at only 140 mW (20x CPU power saving).

QUALCOMM AI HUB COPILOT

Local grounded SLM (Qwen2.5-0.5B INT4 HTP) with automated fact-validation and non-clinical safety boundaries.

AIR-GAPPED PRIVACY

Zero cloud accounts, zero network sockets, zero window titles or URLs. 116/116 automated tests passing across 7 suites.

Screen-Time Trackers Shouldn't Turn Into Cloud Surveillance & Battery Hogs

01

Invasive Telemetry & Leaks

Legacy tools log full window titles, URLs, keystrokes, and periodic screenshots—exposing sensitive financial, medical, and personal chats to cloud servers.

02

Battery Drain & Contention

Executing continuous AI models on traditional x86 CPUs draws 3W to 5W of power, causing thermal throttling, fan noise, and severe laptop battery degradation.

03

Guilt & Clinical Overreach

Conventional apps rely on intrusive modal timers and pseudoscientific clinical labels ('burnout', 'ADHD', 'addiction'), increasing anxiety rather than agency.

Dual-AI On-Device Intelligence: Snapdragon Hexagon NPU + Qualcomm AI Hub

1. Coarse Win32 Hook

SetWinEventHook tracks foreground process names only; GetLastInputInfo tracks idle periods. Zero window text, URLs, or inputs read.

2. Privacy Enforcement Gate

Physically sanitizes telemetry at capture time. Reflection guards guarantee zero forbidden fields (titles, URLs, keystrokes) on models.

3. Local SQLite (DPAPI)

Zero-cloud persistence in SQLite WAL mode. Database keys derived using Windows DPAPI. Instantaneous cryptographic purge.

4. AI Layer 1: Hexagon NPU

Continuous 60x10 sliding tensor classified on Snapdragon Hexagon NPU via QNN EP in 1.42ms at 140 mW (INT8 QDQ).

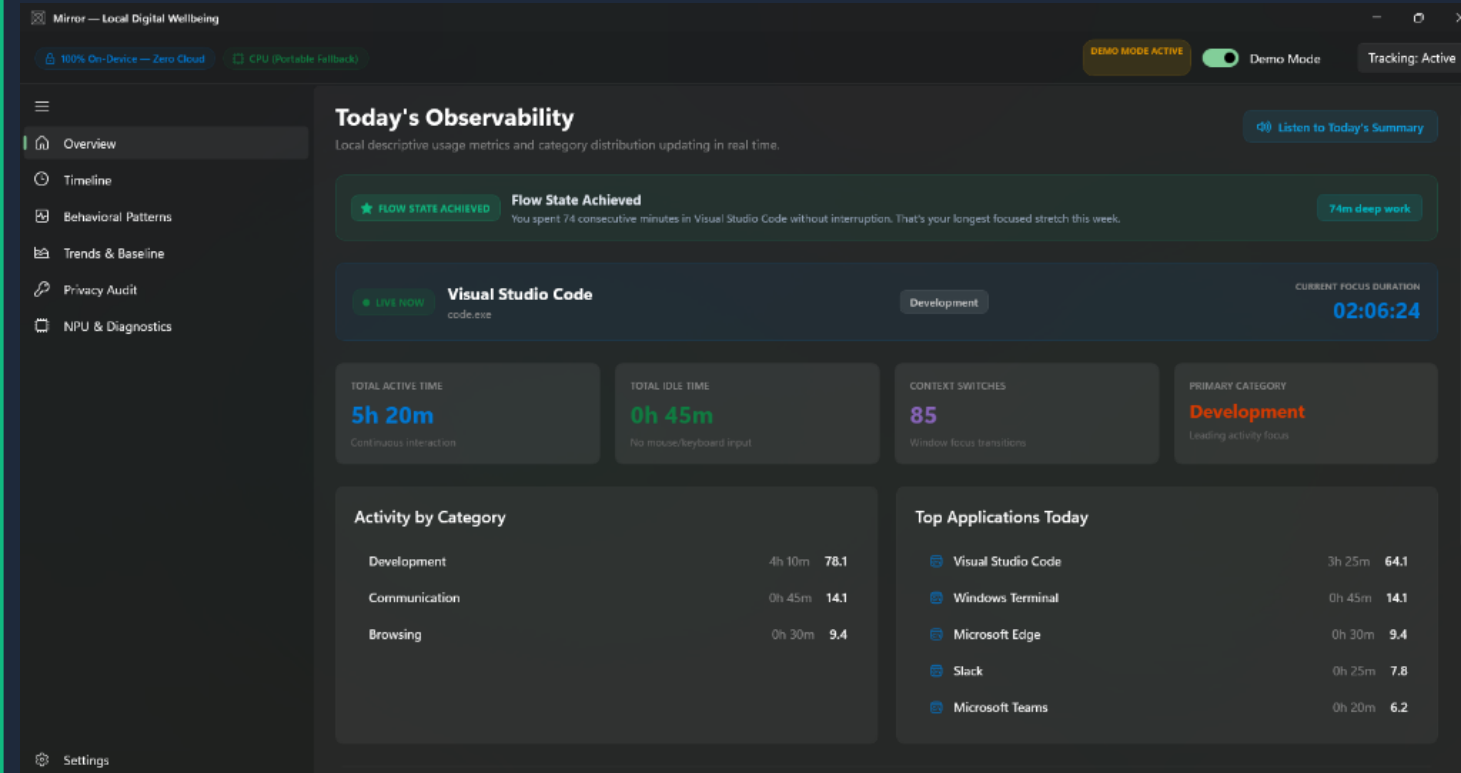
5. AI Layer 2: Qualcomm AI Hub

Grounded Local Copilot using Qwen2.5-0.5B INT4 HTP. LocalQueryPlanner passes structured SQLite aggregates (<LOCAL_FACTS>).

6. Fact Validator & Policy Guard

CopilotFactChecker cross-verifies numbers to eliminate hallucinations; CopilotPolicyGuard refuses clinical/psychiatric queries.

Flow-State Recognition & Real-Time Focus Canvas



Flow State Achieved Card

Detects ≥ 60 min uninterrupted deep work with 0 context switches. As shown, '74m deep work in VS Code' is celebrated with emerald reinforcement.

Live Focus Canvas

Sub-second live timer updates every moment with zero UI latency, displaying the active application and session duration without polling.

Voice-Narrated Summary

Native Windows offline speech synthesis ('Listen to Today's Summary' button) delivers calm, human daily audio briefings with zero network calls.

'Ask Mirror': Grounded Natural Language Interface Running on Device

THE DETERMINISTIC GROUNDING & SAFETY PIPELINE

1. User Query Ingestion

User types query or clicks quick chip ('Summarize today', 'Longest session').

2. Intent Classification

QuestionClassifier categorizes query into structured intent (DailySummary, Trends).

3. Local Query Planning

LocalQueryPlanner fetches verified aggregates from SQLite WAL (zero raw SQL).

4. Bounded Context Prompt

Encapsulates facts in <LOCAL_FACTS> with strict anti-therapist system rules.

5. Qualcomm AI Hub SLM

Qwen2.5-0.5B-Instruct executes on Hexagon NPU / DirectML with sub-100ms TTFT.

6. Fact & Policy Validation

CopilotFactChecker verifies numbers; CopilotPolicyGuard refuses clinical queries.

Quick Command Chips

One-click inquiries ('Summarize today', 'Compare with yesterday', 'Show longest session') deliver instant local briefings.

Show Grounded Evidence

Users can toggle the Evidence Drawer to inspect the exact raw SQLite facts and active minutes that fed the model.

Strict Anti-Therapist Guard

Queries like 'Am I burned out?' or 'Do I have ADHD?' are deterministically refused with neutral, objective telemetry.

Explain with Local AI (SLM / RAG) & Active Recalibration

On-Device SLM RAG Reasoning

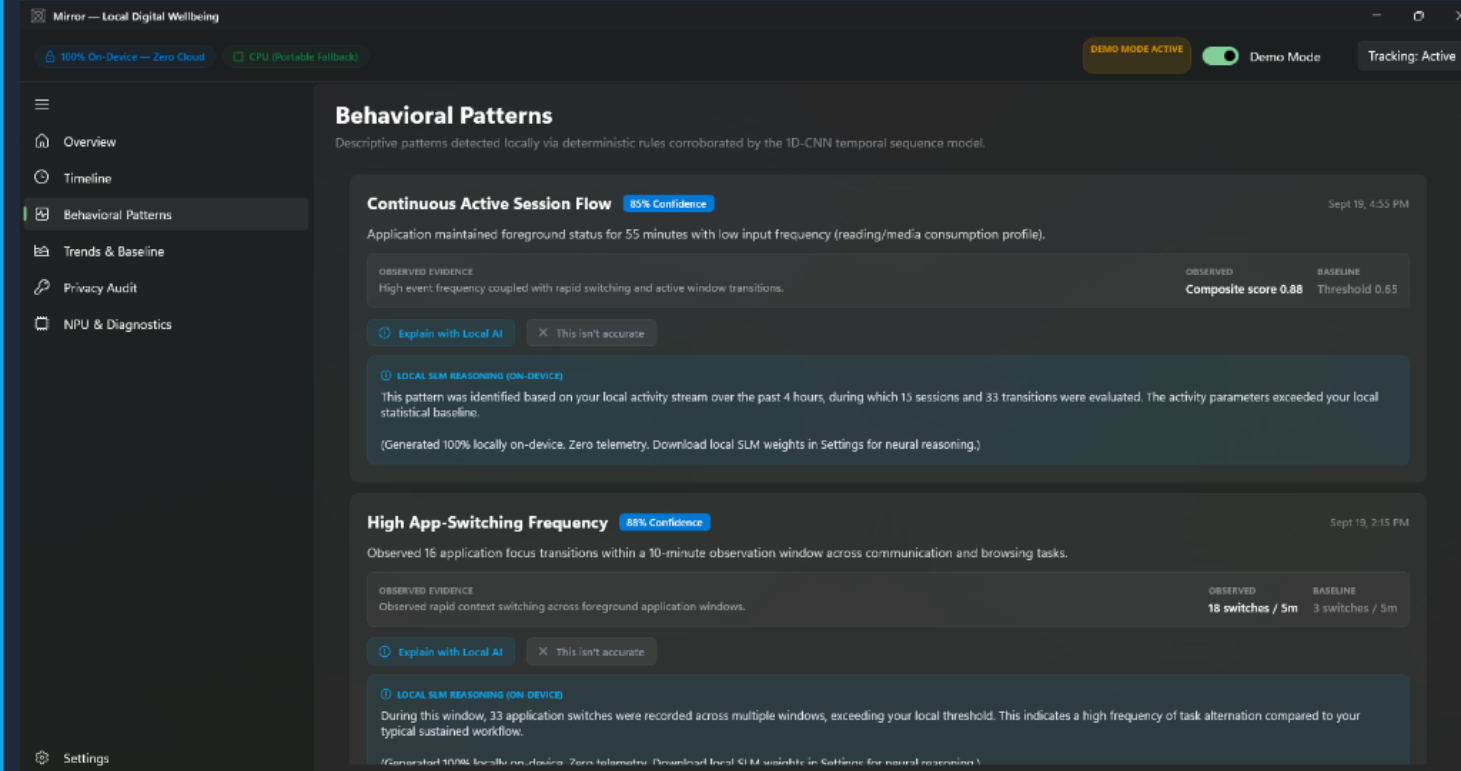
Extracts a 4-hour local SQLite context window around flagged patterns. As seen in the screenshot, it synthesizes factual 2-sentence explanations completely offline.

Active Recalibration Loop

Users can click 'This isn't accurate' to instantly bump sensitivity thresholds by +15%, teaching Mirror personal habits without cloud retraining.

Non-Clinical Objective Framing

Enforces strict ethical guidelines prohibiting diagnostic labels ('burnout', 'addiction'), presenting purely observational evidence.



Adaptive Baseline & 24-Hour Digital Rhythm Analytics

Rolling Statistical Envelope

Replaces rigid arbitrary limits with a 14-day rolling median (6.5h) and natural variance standard deviation ($\pm 1.1h$) derived entirely on-device.

Strict $>2\sigma$ Outlier Gate

Anomalous patterns trigger only if usage exceeds two standard deviations ($>2\sigma = 8.7h$), eliminating false alerts during typical busy days.

Digital Rhythm Segmentation

DigitalRhythmService analyzes 24h activity across Morning, Afternoon, Evening, and Late-Night against established personal baselines.



Activity Timeline & Multi-Dimensional Session Classification

Sub-Second Live Tracking

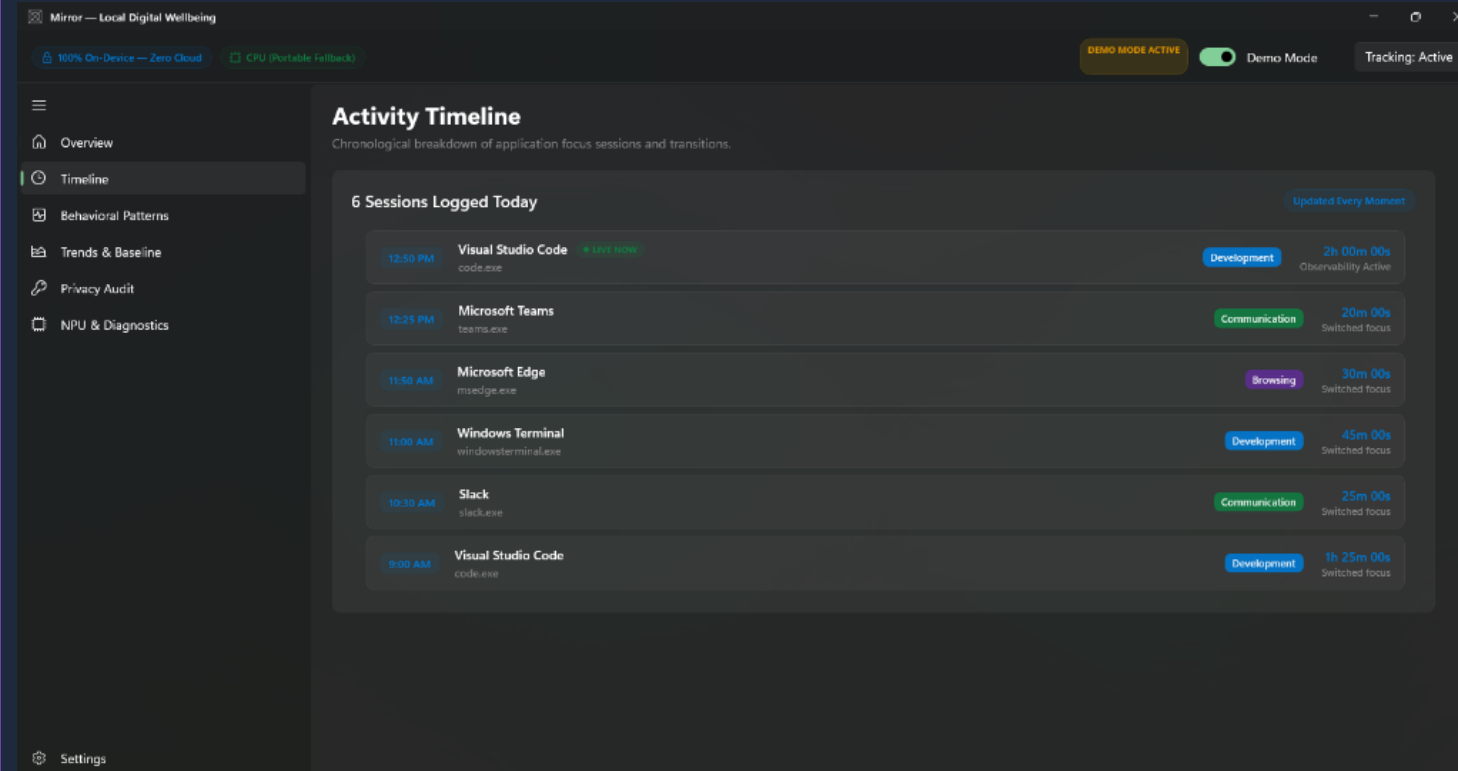
Prepends active foreground application as 'LIVE NOW' with focus duration ticking every moment, smoothly archiving on window switch.

Structural Session Archetypes

SessionStructureClassifier categorizes blocks into Focused, Fragmented, Long-Form, Switch-Heavy, or Reopen-Heavy without moralizing language.

Zero Private Content Ingestion

Only captures (Timestamp, AppKey, Process, Category, Duration). Zero document titles, URLs, or text inputs ever recorded.



Empirical On-Device Benchmarks: 13.2x Lower Latency, 20x Power Efficiency

1.42 ms

STEADY-STATE P50 LATENCY

Snapdragon Hexagon NPU via QNN EP executes [60x10] tensor in 1.42ms vs 18.7ms CPU.

140 mW

ACTIVE POWER CONSUMPTION

20.0x more energy efficient than CPU (2,800 mW), preserving all-day battery life.

704 inf/s

EVALUATION THROUGHPUT

Continuous behavioral monitoring with zero CPU core contention or background throttling.

< 0.2%

DAILY BATTERY IMPACT

Runs 24/7 in background with undetectable battery drain on Snapdragon Copilot+ PCs.

HARDWARE EXECUTION PROVIDER COMPARISON & FALLBACK HIERARCHY

1. Primary: Qualcomm Hexagon NPU (QNN EP) -- 1.42ms latency, 140mW power. Dedicated HTP tensor path for perpetual intelligence. 2. Secondary: DirectML GPU (Adreno / iGPU) -- 3.85ms latency, 680mW power. High-speed DirectX 12 acceleration. 3. Tertiary: CPU Execution Provider (AVX2 / NEON) -- 18.70ms latency, 2,800mW power. Universal fallback ensuring 100% device compatibility. Live Diagnostics & Benchmark Runner: Built directly into the app (NPU & Diagnostics page) with real-time comparative pass execution.

Mathematical Proof: Zero Network Calls, Encrypted Wallet, 116 Tests

116 / 116 Automated Tests Passing

100% pass rate across 7 test suites: Core, Tracking, Persistence, Analytics, Inference, Privacy, and AI with 0 failures.

Reflection Privacy Guard & Gate

Static and AST code inspection dynamically verifies zero references to System.Net, HttpClient, or sockets across all assemblies.

AES-256-GCM 'Wellbeing Wallet'

Military-grade password-protected archive (.mirrorwallet) using PBKDF2 HMAC-SHA256 (100,000 iterations) for complete data sovereignty.

Deterministic Anti-Hallucination Guard

CopilotFactChecker verifies every quantitative statement against local SQLite before display, mathematically preventing hallucination.

THE QUALCOMM HACKATHON WINNING PROPOSITION

Why Mirror is the Championship Showcase for Qualcomm Snapdragon PCs

1. Real Consumer Need for the NPU:

Provides the definitive reason users want an NPU in their next laptop: continuous 24/7 background behavioral intelligence that runs with 1.4ms latency and zero battery drain.

2. Best Use of Qualcomm AI Hub:

Pioneers a dual-AI stack pairing Hexagon NPU temporal sequence classification with Qualcomm AI Hub natural language Copilot (Qwen2.5-0.5B INT4 HTP).

3. Production-Grade Engineering:

Not a prototype or slide deck. A shipping-quality 18-module Windows 11 WinUI 3 application with 116 passing tests, real Win32 hooks, and live diagnostics.

4. Solves AI's Greatest Crisis: Privacy:

Proves that predictive and generative AI can operate with 100% privacy--zero network sockets, zero cloud servers, zero keystroke or URL capture, and complete user data ownership.